

Unit 14: Tectonic Force — Stress, Structures & Earthquakes

Reading: pages 199–211 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Which type of stress is most common at convergent plate boundaries?
 - a) Tension
 - b) Shear
 - c) Confining stress
 - d) Compression
2. When a rock under stress does *not* return to its original shape after the stress is removed, it has undergone
 - a) plastic deformation
 - b) fracture
 - c) confining stress
 - d) elastic deformation
3. In a fold where the oldest rocks sit at the center and the layers arch upward and dip away from that center, the structure is
 - a) a syncline
 - b) an anticline
 - c) a monocline
 - d) a joint
4. In a *normal* fault,
 - a) the hanging wall drops down relative to the footwall
 - b) the footwall drops down relative to the hanging wall
 - c) the two blocks slide horizontally past each other
 - d) there is fracturing but no movement
5. About what share of all earthquakes strike around the Pacific Ocean basin, the Ring of Fire?
 - a) About 15%
 - b) About 50%
 - c) About 80%
 - d) About 95%

6. On the Richter and moment magnitude scales, an increase of one whole number means the amplitude of the largest seismic wave increases by a factor of
- a) 2
 - b) 10
 - c) 30
 - d) 100

Part B · True or False

2 questions × 1 point = 2 points

7. The *focus* of an earthquake is the point on the land surface directly above where the rocks rupture.
TRUE / FALSE
8. More people are typically killed by the fires that follow an earthquake than by the earthquake itself.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between the Mercalli Intensity scale and the moment magnitude scale. Which do most seismologists use today, and why?

10. What is *liquefaction*, and why does it make Salt Lake City and the Wasatch Front especially vulnerable?

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